



NEXUS 2026 · FINAL ROUND

Investor Portfolio Monitoring & Risk Management System

See your risk clearly.

Team Yash Mandaliya · Kaivan Parikh

M.Sc. Information Technology · GLS University, Ahmedabad

THE PROBLEM

Problem Statement

Individual investors hold assets across multiple platforms, resulting in fragmented information and limited visibility into their overall financial position. Without a centralized system, tracking portfolio performance, assessing diversification, monitoring risk exposure, and identifying trends becomes difficult.



Fragmented

Holdings scattered across brokers — no single view.



Risk is invisible

Apps show profit, never downside exposure.



No insight

Diversification and trends go unnoticed.

Build a system that consolidates portfolio data, monitors performance, analyzes asset allocation, and generates risk insights.

OUR SOLUTION

Clarisk — a unified risk dashboard

A single web platform where an investor signs in by phone, consolidates holdings from any platform, and instantly sees a live valuation, a full quantitative risk profile, a rebalancing suggestion, and real-time alerts — refreshing itself automatically.

1

Consolidate

Sign in by phone; add holdings by hand, CSV, or simulated broker sync — one unified portfolio.

2

Analyse

Live valuation, allocation, and a full risk engine — VaR, beta, Sharpe, drawdown, correlation.

3

Act

Rebalancing suggestion, live breach alerts, and a downloadable PDF risk report.

Working product — not a mockup. Backed by 550+ automated tests.

COVERAGE

Every requirement, delivered

REQUIREMENT	HOW CLARISK DELIVERS IT	STATUS
Consolidates portfolio data	Per-user portfolio; add by hand, CSV, or broker sync	Done
Multi-platform aggregation	Simulated broker connect (Zerodha/Groww/Upstox/ICICI) + CSV	Demo
Monitors performance	Live valuation, P/L, value & drawdown charts	Done
Analyses asset allocation	Allocation by holding and sector	Done
Risk exposure metrics	VaR (3 methods), CVaR, beta, volatility, Sharpe, Sortino, drawdown	Done
Diversification insights	Correlation matrix + concentration (HHI)	Done
Identifies trends	Performance curve & underwater drawdown over time	Done
Reporting tools	One-click PDF risk report	Done

LIVE PRODUCT

A real, working platform

 **Clarisk**
See your risk clearly.

Dark

Sign in

Welcome back. Send a fresh code to the number below to sign in again.

Development mode. App verification is off, so there is no reCAPTCHA at all — but only a Firebase *test* number will sign in, with its fixed code. No SMS is sent.

PHONE NUMBER

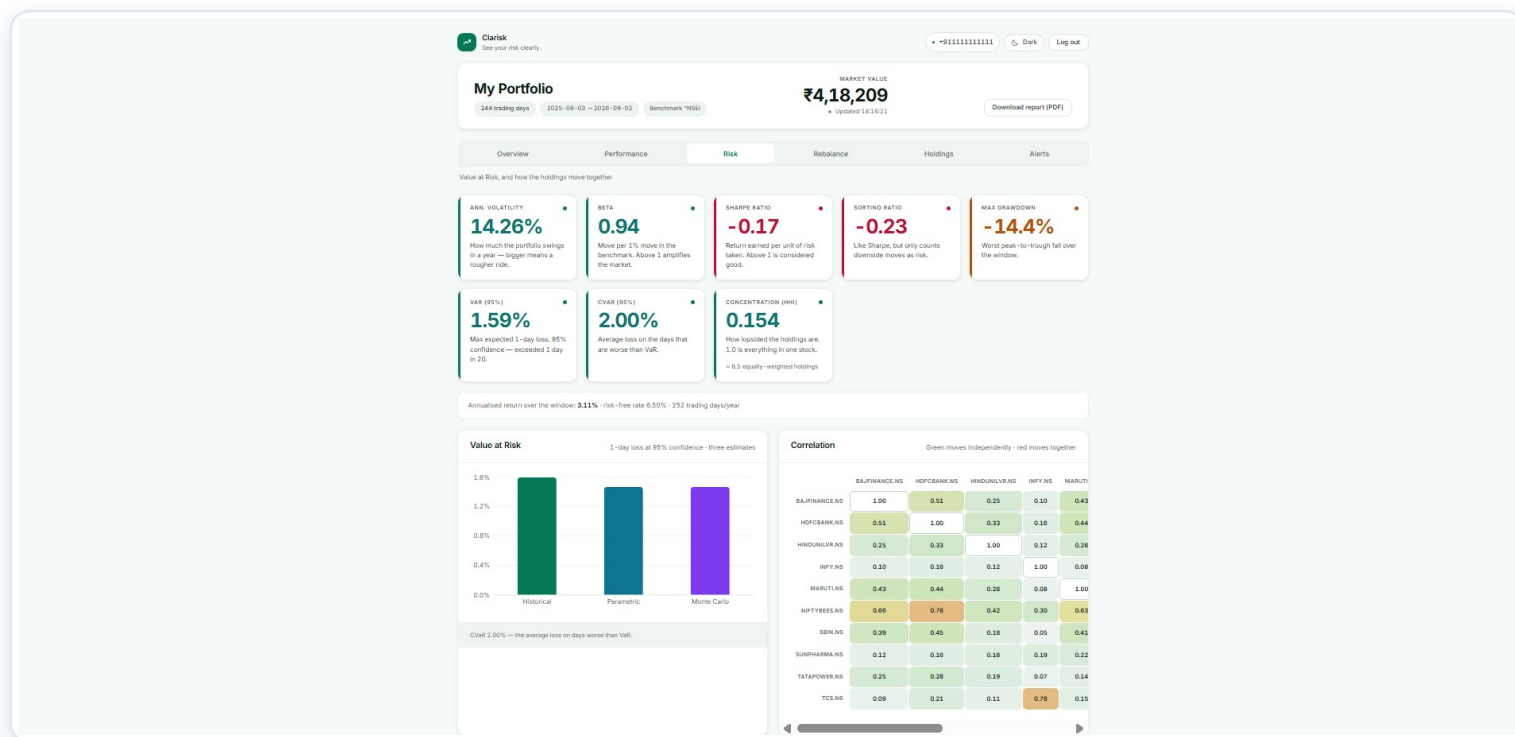
+911111111111

Send OTP

Not you? [Use a different number](#)

reCAPTCHA is mocked in development. No SMS is sent to test numbers.

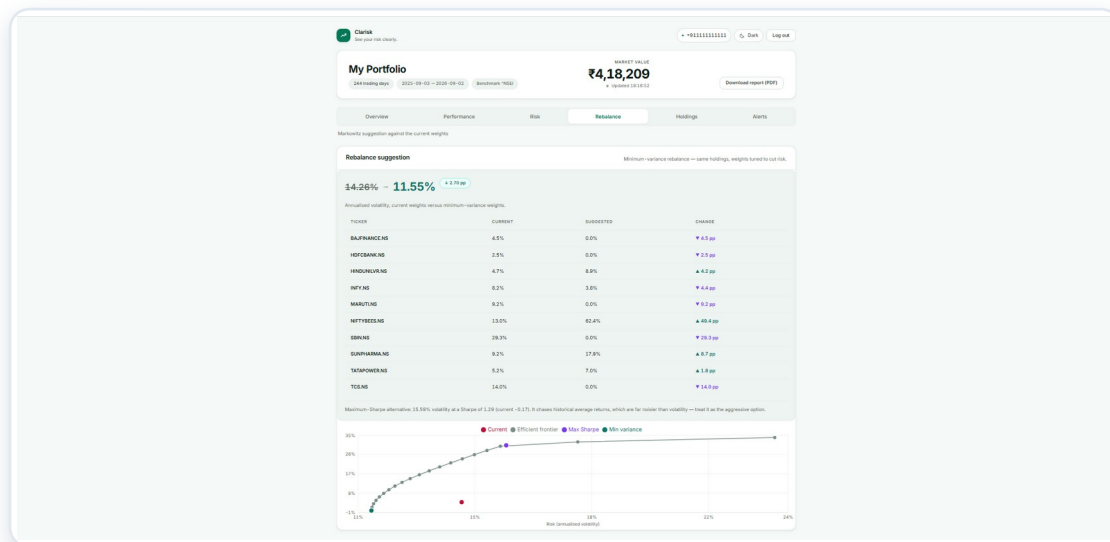
Phone-OTP sign in



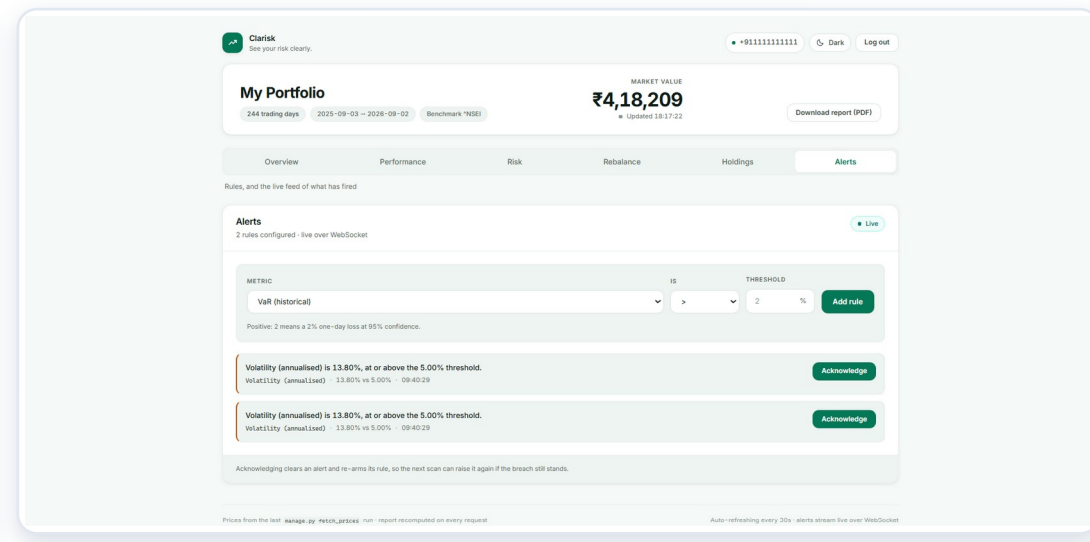
Risk dashboard — VaR, beta, Sharpe, drawdown, correlation, all computed live from market data

LIVE PRODUCT

Rebalancing & real-time alerts



Markowitz rebalance + efficient frontier — cuts volatility 14.3% → 11.6%



Live risk alerts pushed over WebSocket the moment a limit breaks

₹4.18L

Live portfolio value

60 sec

Auto-refresh cycle

Live

WebSocket alerts

MULTI-PLATFORM AGGREGATION

Consolidating across platforms

OverviewPerformanceRiskRebalanceHoldingsAlerts

Every position, and the forms that change them

Manage holdings

Connect a broker, add a position by hand, or import a CSV. Editing costs ₹9 per round — you are asked when you save, not before.

Connect a broker

Simulated

Z

Zerodha

Import sample

G

Groww

Import sample

U

Upstox

Import sample

I

ICICI Direct

Import sample

Demo: broker sync is simulated with sample data. Live API integration (e.g. Zerodha Kite Connect) is on our roadmap.

Add one

TICKER

QUANTITY

AVG BUY PRICE

BUY DATE optional

ASSET TYPE optional

SECTOR optional

RELIANCE.NS

10

1400.50

dd-mm-yyyy

Equity

Energy

Add holding

Import a CSV

Required columns: `ticker` `quantity` `avg_buy_price` . Optional: `buy_date` `asset_type` `sector` . Up to 500 rows, 1 MB. [Download sample.csv](#)

Choose File

No file chosen

Import

Honest by design

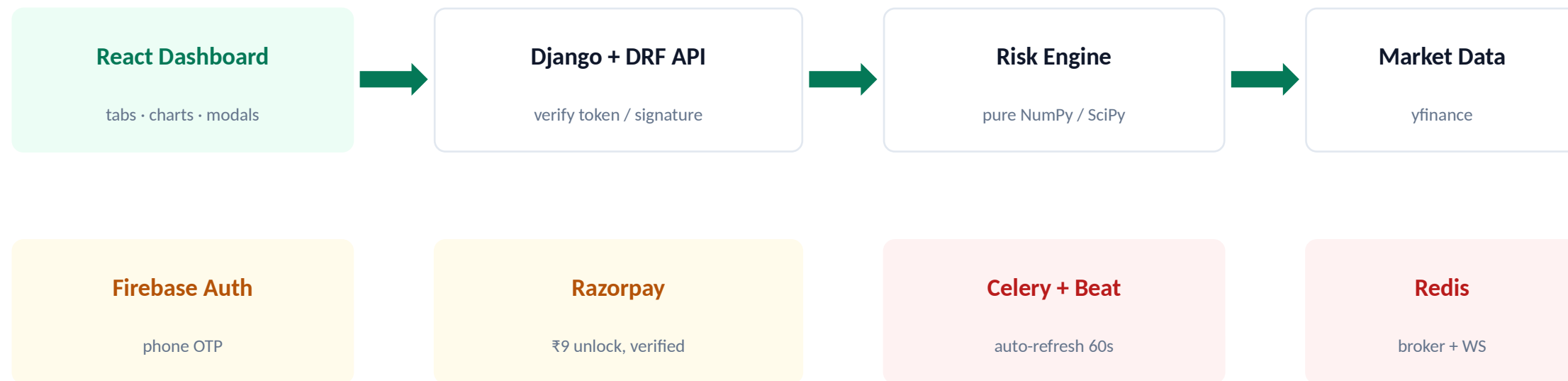
Today — holdings consolidate from any platform via CSV export + our simulated broker connect.

Next — live broker API sync (Zerodha Kite Connect). Our swappable data-provider layer already supports it — the seam is built.

Connect Zerodha, Groww, Upstox or ICICI — holdings flow into one portfolio; overlapping stocks consolidate automatically.

ARCHITECTURE

How it works



Data flow

Sign in (OTP) → add / import holdings (₹9 gate) → fetch live + 252-day history → align returns → risk engine builds the report → dashboard renders it → Celery re-scans and pushes alerts every 60 seconds.

Institutional-grade risk analytics

Every metric is a pure, independently-tested function — verified against known mathematical results, not assumed.

3

VaR methods

Historical • Parametric • Monte Carlo

7+

Risk metrics

Beta • Sharpe • Sortino • CVaR • Drawdown • HHI

Markowitz

Optimiser

Min-variance + efficient frontier

Why minimum-variance leads

It needs only the covariance matrix. Maximum-Sharpe depends on noisy return forecasts and over-concentrates — so we show it as a labelled footnote, not the recommendation. Knowing that distinction is the difference between using a formula and understanding it.

STACK

Built to be real



Backend

Django · DRF



Risk maths

NumPy · Pandas · SciPy



Market data

yfinance (NSE)



Auth

Firebase Phone OTP



Payments

Razorpay



Real-time

Django Channels · daphne



Scheduling

Celery · Beat · Redis



Frontend

React · Vite · Recharts

SQLite storage · ReportLab PDF · 550+ automated tests, all passing

What's real vs what's next

Working today

- ✓ Phone-OTP auth & per-user portfolios
- ✓ Live valuation & full risk engine
- ✓ Rebalancing + efficient frontier
- ✓ Real-time WebSocket alerts
- ✓ Auto-refresh every 60 seconds
- ✓ CSV import & PDF report export
- ✓ Razorpay ₹9 payment gate
- ✓ Simulated multi-broker sync

On the roadmap

- Live broker API sync (Kite Connect)
- Real SMS OTP + production reCAPTCHA
- Multiple portfolios per user
- Stress testing / scenario analysis
- Multi-currency & FX risk
- Mobile app + scheduled email reports

Why Clarisk stands out

- ✓ **A working product** — Not slides — a running platform with live market data, auth and payments.
- ✓ **Real quant depth** — VaR, CVaR, Markowitz — verified against known results, 550+ tests.
- ✓ **Honest & adaptable** — We name what's simulated, and the architecture is built to extend fast.

See your risk clearly.